Technische Hochschule Würzburg-Schweinfurt
Faculty of Computer Science and Business Information Systems

Master's Thesis

Title of the Thesis

Submitted to the Technische Hochschule Würzburg-Schweinfurt, Faculty of Computer Science and Business Information Systems, in partial fulfilment of the requirements for the degree of Master of Science in Artificial Intelligence

Harsha

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First examiner: Prof. Dr. A B
Second examiner: Prof. Dr. C D

Written in cooperation with BMW Group

[1]

Abstract

Write last. About 200–300 words: context, problem, approach, key results, conclusion.

Acknowledgements

TODO

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List of Acronyms

8D Eight Disciplines Problem Solving

AI Artificial Intelligence

API Application Programming Interface

EM Exact Match

FMEA Failure Mode and Effects Analysis

KG Knowledge Graph

LLM Large Language Model

NL Natural Language

QMT Quality Management (BMW Group)

RAG Retrieval-Augmented Generation

RCA Root Cause Analysis

RPN Risk Priority Number

SQL Structured Query Language

1 Introduction

Short opening: why quality engineers need to query 8D and FMEA knowledge, and why that is hard today.

1.1 Motivation

Example of acronyms: a Large Language Model (LLM) translates a question in Natural Language (NL) into a query on a Knowledge Graph (KG). The second time, LLM and KG appear in short form only. Plural: LLMs.

Example of a figure reference: figure 1.1 shows ... Example of a citation [1].

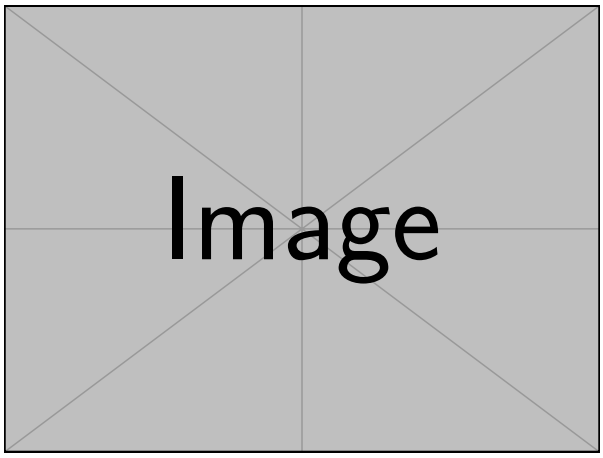


Figure 1.1: Placeholder figure.

1.2 Research Questions

RQ1 TODO

RQ2 TODO

RQ3 TODO

1.3 Contributions

TODO

1.4 Structure of this Thesis

Chapter 2 introduces ... Chapter 3 describes ... Chapter 4 presents ... Chapter 5 evaluates ... Chapter 6 concludes ...

2 Fundamentals

TODO: one paragraph saying what this chapter covers.

2.1 Quality Management with 8D and FMEA

2.1.1 The 8D Method

TODO

2.1.2 Failure Mode and Effects Analysis

TODO

2.2 Knowledge Graphs and Neo4j

TODO

2.3 Large Language Models and Text-to-Query

TODO

3 Problem Statement

TODO

3.1 Current Situation

TODO

3.2 Requirements

ID	Type	Requirement
R1	Functional	TODO
R2	Functional	TODO
R3	Non-functional	TODO

Table 3.1: Requirements for the system.

4 Solution

TODO

4.1 Graph Schema

Listing 4.1 shows an example Cypher query.

```
// Find failure modes linked to a component
MATCH (c:Component {name: $component})<-[:AFFECTS]-(f:FailureMode)
OPTIONAL MATCH (f)-[:CAUSED_BY]->(cause:Cause)
RETURN f.description AS failureMode,
       collect(DISTINCT cause.description) AS causes
ORDER BY failureMode
LIMIT 10
```

Listing 4.1: Example Cypher query (placeholder).

4.2 Agent Architecture

Listing 4.2 shows an example in Python.

```
def run_query(driver, cypher: str, params: dict) -> list[dict]:
    """Execute a Cypher query and return the records."""
    with driver.session() as session:
        result = session.run(cypher, params)
        return [record.data() for record in result]
```

Listing 4.2: Example Python code (placeholder).

A prompt can be shown with the `prompt` style:

```
You are an expert in Neo4j Cypher. Given the schema below,
translate the user's question into a single Cypher query.
```

Listing 4.3: Example prompt (placeholder).

5 Evaluation

TODO

5.1 Experimental Setup

TODO

5.2 Results

Table 5.1 shows how a results table can look.

Configuration	Accuracy	Valid queries	Latency (s)
Baseline	0.00	0.00	0.0
Variant A	0.00	0.00	0.0
Variant B	**0.00**	**0.00**	0.0

Table 5.1: Placeholder results table.

5.3 Discussion

TODO

6 Conclusion

TODO

6.1 Summary

TODO

6.2 Future Work

TODO

Appendix

Appendix A: Example Table

Heading 1	Heading 2
Title 1	Description 1
Title 2	Description 2

Table 1: Placeholder appendix table.

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Bibliography

[1] E. Gamma et al. *Entwurfsmuster: Elemente wiederverwendbarer objektorientierter Software*. Programmer's choice. Pearson Deutschland, 2011.

Statutory Declaration

I hereby declare that I have written this Master's thesis independently and have not submitted it elsewhere for examination purposes. All sources and aids used have been indicated, and all direct and indirect quotations have been marked as such.

Harsha, October 12, 2026

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